Name
Date
Course

The Chain Rule

Differentiation Formulas

Constant Rule i) $\frac{d}{dx}(c) = 0$ "C" is a constant.

Power Rule ii) $\frac{d}{dx}(x^n) = nx^{n-1}$ "n" is a constant.

Constant Multiple Rule iii) $\frac{d}{dx}[cf(x)] = cf'(x)$ "C" is a constant.

Sum Rule iv) $\frac{d}{dx}[f(x)+g(x)] = f'(x) + g'(x)$

Difference Rule v) $\frac{d}{dx}[f(x)-g(x)] = f'(x) - g'(x)$

Product Rule vi) $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$

Quotient Rule vii) $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$

Trigonometric Differentiation Formulas

I) $\frac{d}{dx}(\sin x) = \cos x$

II) $\frac{d}{dx}(\cos x) = -\sin x$

III) $\frac{d}{dx}(\csc x) = -\csc x \cot x$

IV) $\frac{d}{dx}(\sec x) = \sec x \tan x$

V) $\frac{d}{dx}(\tan x) = \sec^2 x$

VI) $\frac{d}{dx}(\cot x) = -\csc^2 x$

The Chain Rule viii) If $h(x) = f(g(x))$, then

$$\boxed{h'(x) = f'(g(x))\, g'(x)}$$

Three Steps for the Chain Rule

1) Find the derivative of the outside function while
2) leaving the inside function unchanged.
3) Multiply the resulting expression by the derivative of the inside function.

Practice

Find $\frac{dy}{dx}$.

① $y = (x^4 - 3x + 7)^3$

$$\frac{dy}{dx} = 3(x^4-3x+7)^{3-1}\frac{d}{dx}(x^4-3x+7)$$

$$= \boxed{3(x^4-3x+7)^2(4x^3-3)}$$

② $y = \sin(x^2)$

$$\frac{dy}{dx} = \cos(x^2)\frac{d}{dx}(x^2) = [\cos(x^2)](2x) = \boxed{2x\cos(x^2)}$$

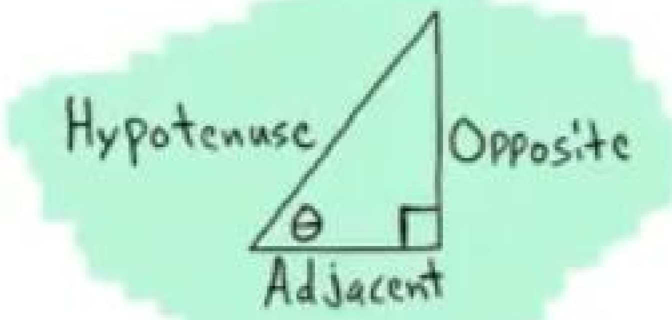
③ $y = \sin^2 x$

$y = (\sin x)^2$

$$\frac{dy}{dx} = 2(\sin x)^{2-1}\frac{d}{dx}(\sin x) = 2(\sin x)^1 \cos x$$

$$= 2\sin x \cos x$$

$$= \boxed{\sin(2x)}$$



$\sin\theta = \frac{Opp}{Hyp}$ $\quad\csc\theta = \frac{Hyp}{Opp}$

$\cos\theta = \frac{Adj}{Hyp}$ $\quad\sec\theta = \frac{Hyp}{Adj}$

$\tan\theta = \frac{Opp}{Adj}$ $\quad\cot\theta = \frac{Adj}{Opp}$

Opposite Angle Identities

$\sin(-\theta) = -\sin\theta$ $\quad\cos(-\theta) = \cos\theta$ $\quad\tan(-\theta) = -\tan\theta$

$\csc(-\theta) = -\csc\theta$ $\quad\sec(-\theta) = \sec\theta$ $\quad\cot(-\theta) = -\cot\theta$

Complete Rotation and Half Rotation Identities

$\sin(\theta \pm 360°) = \sin\theta$ $\quad\sin(\theta \pm 180°) = -\sin\theta$

$\cos(\theta \pm 360°) = \cos\theta$ $\quad\cos(\theta \pm 180°) = -\cos\theta$

$\tan(\theta \pm 360°) = \tan\theta$ $\quad\csc(\theta \pm 180°) = -\csc\theta$

$\csc(\theta \pm 360°) = \csc\theta$ $\quad\sec(\theta \pm 180°) = -\sec\theta$

$\sec(\theta \pm 360°) = \sec\theta$ $\quad\tan(\theta \pm 180°) = \tan\theta$

$\cot(\theta \pm 360°) = \cot\theta$ $\quad\cot(\theta \pm 180°) = \cot\theta$

Complete Rotation Angle Pair Identities and Supplementary Angle Identities

$\sin(360° - \theta) = -\sin\theta$ $\quad\sin(180° - \theta) = \sin\theta$

$\cos(360° - \theta) = \cos\theta$ $\quad\csc(180° - \theta) = \csc\theta$

$\tan(360° - \theta) = -\tan\theta$ $\quad\cos(180° - \theta) = -\cos\theta$

$\csc(360° - \theta) = -\csc\theta$ $\quad\sec(180° - \theta) = -\sec\theta$

$\sec(360° - \theta) = \sec\theta$ $\quad\tan(180° - \theta) = -\tan\theta$

$\cot(360° - \theta) = -\cot\theta$ $\quad\cot(180° - \theta) = -\cot\theta$

Sum to Product Formulas

$\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$

$\sin A - \sin B = 2\sin\left(\frac{A-B}{2}\right)\cos\left(\frac{A+B}{2}\right)$

$\cos A + \cos B = 2\cos\left(\frac{A-B}{2}\right)\cos\left(\frac{A+B}{2}\right)$

$\cos A - \cos B = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$

Product to Sum Formulas

$\sin A\cos B = \frac{1}{2}\left[\sin(A+B) + \sin(A-B)\right]$

$\cos A\sin B = \frac{1}{2}\left[\sin(A+B) - \sin(A-B)\right]$

$\cos A\cos B = \frac{1}{2}\left[\cos(A+B) + \cos(A-B)\right]$

$\sin A\sin B = \frac{1}{2}\left[\cos(A-B) - \cos(A+B)\right]$

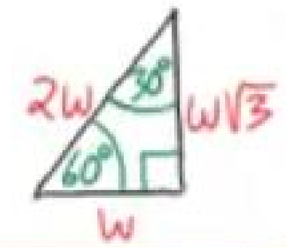
The Area of an SAS Triangle

The area of a triangle is equal to the length of one side times the length of another side times the sine of the measure of the angle between the two sides divided by two.

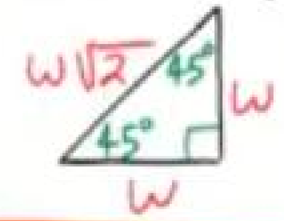
$Area = \frac{1}{2}ab\sin C$ $\quad Area = \frac{1}{2}bc\sin A$

$Area = \frac{1}{2}ac\sin B$

30°-60°-90° Triangle Theorem: If the inner angles of a triangle are 30°, 60°, and 90°, and the length of the side opposite the 30° angle is W, then the length of the hypotenuse is 2W and the length of the side opposite the 60° angle is $W\sqrt{3}$.



45°-45°-90° Triangle Theorem: If the inner angles of a triangle are 45°, 45°, and 90°, and the length of the two sides opposite the 45° angles are W, then the length of the hypotenuse is $W\sqrt{2}$.



Quarter Rotation Identities

$\sin(\theta \pm 90°) = \pm\cos\theta$ $\quad\cos(\theta \pm 90°) = \mp\sin\theta$

$\csc(\theta \pm 90°) = \pm\sec\theta$ $\quad\sec(\theta \pm 90°) = \mp\csc\theta$

$\tan(\theta \pm 90°) = -\cot\theta$

$\cot(\theta \pm 90°) = -\tan\theta$

Complementary Angle Identities

$\sin(90° - \theta) = \cos\theta$ $\quad\csc(90° - \theta) = \sec\theta$

$\cos(90° - \theta) = \sin\theta$ $\quad\sec(90° - \theta) = \csc\theta$

$\tan(90° - \theta) = \cot\theta$

$\cot(90° - \theta) = \tan\theta$

Scenarios when Solving Triangles

The Law of Sines

AAS $\quad$ ASA $\quad$ SSA

$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

SSA → Ambiguity
i) No triangle
ii) One triangle
iii) Two triangles

The Law of Cosines

SSS $\quad$ SAS

$c^2 = a^2 + b^2 - 2ab\cos C$

$b^2 = c^2 + a^2 - 2ac\cos B$

$a^2 = b^2 + c^2 - 2bc\cos A$

Heron's Formula

If a, b, and c are the side lengths of a triangle, then the area of the triangle is equal to the expression below.

$Area = \sqrt{s(s-a)(s-b)(s-c)}$

where $s = \frac{a+b+c}{2}$

Reciprocal Identities

$\sin\theta = \frac{1}{\csc\theta}$ $\quad\csc\theta = \frac{1}{\sin\theta}$

$\cos\theta = \frac{1}{\sec\theta}$ $\quad\sec\theta = \frac{1}{\cos\theta}$

$\tan\theta = \frac{1}{\cot\theta}$ $\quad\cot\theta = \frac{1}{\tan\theta}$

Ratio Identities

$\tan\theta = \frac{\sin\theta}{\cos\theta}$

$\cot\theta = \frac{\cos\theta}{\sin\theta}$

Pythagorean Identities

$\sin^2\theta + \cos^2\theta = 1$

$\tan^2\theta + 1 = \sec^2\theta$ $\quad 1 + \cot^2\theta = \csc^2\theta$

Sum and Difference Formulas

$\sin(A \pm B) = \sin A\cos B \pm \sin B\cos A$

$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$

Double Angle Formulas

$\boxed{\sin(2A) = 2\sin A\cos A}$

$\boxed{\cos(2A) = \cos^2 A - \sin^2 A}$

$\boxed{\cos(2A) = 1 - 2\sin^2 A}$

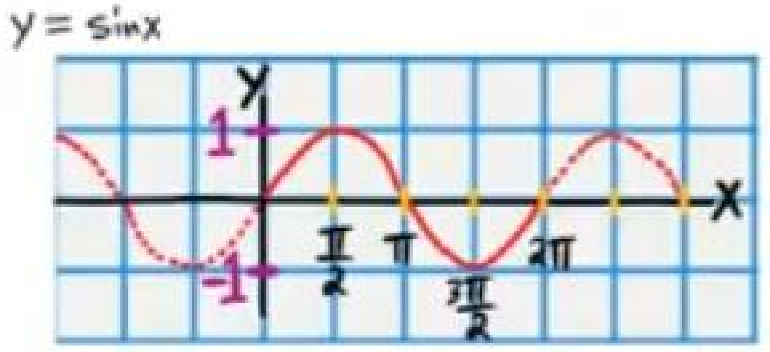
$\boxed{\cos(2A) = 2\cos^2 A - 1}$

Half Angle Formulas

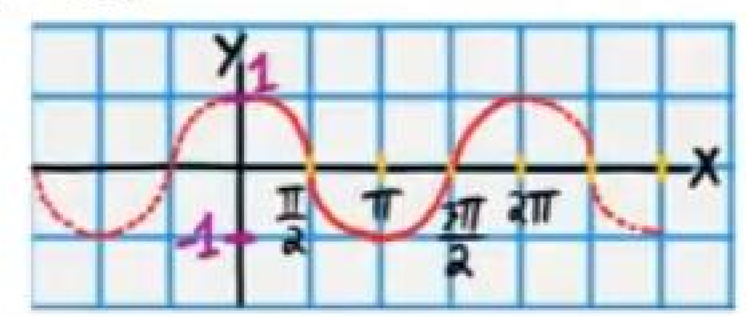
$\sin\left(\frac{A}{2}\right) = \pm\sqrt{\frac{1-\cos A}{2}}$

$\cos\left(\frac{A}{2}\right) = \pm\sqrt{\frac{1+\cos A}{2}}$

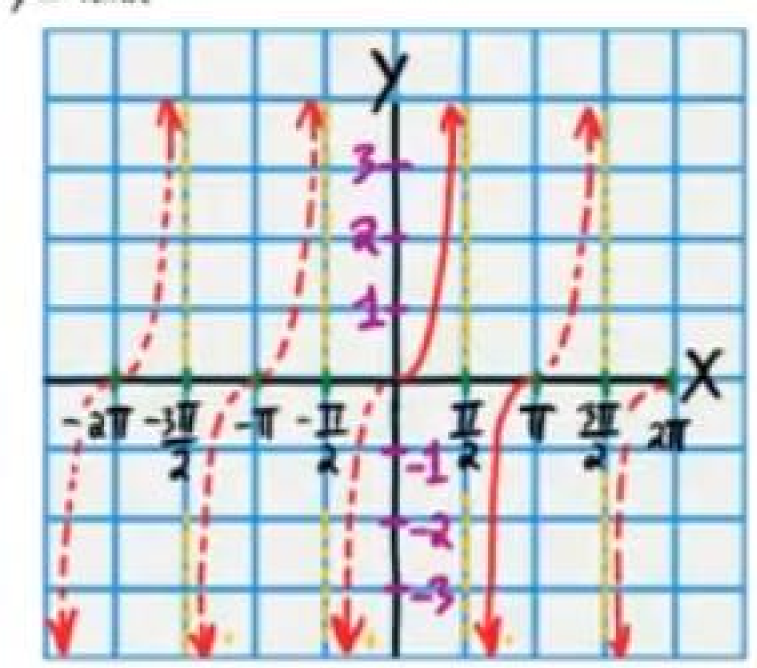
$y = \sin x$



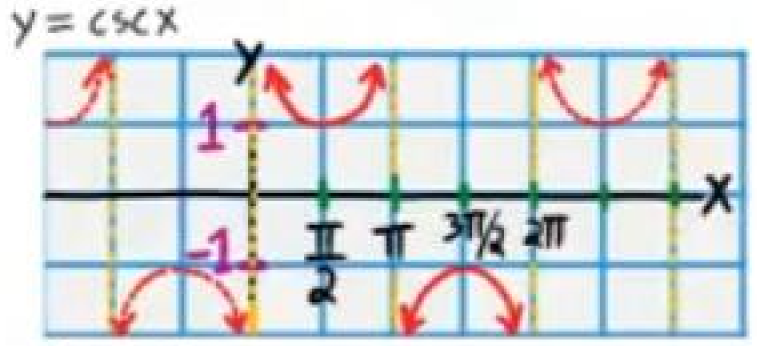
$y = \cos x$



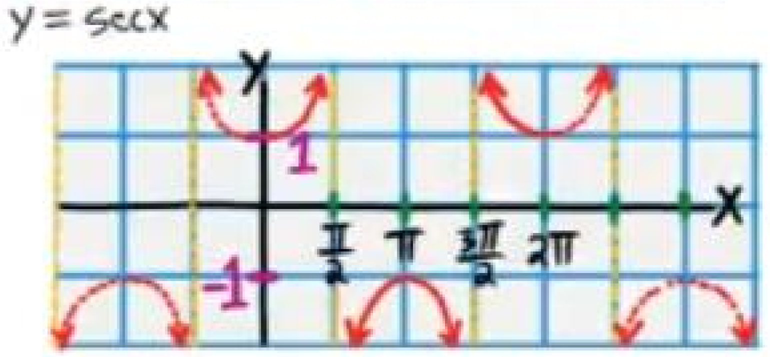
$y = \tan x$



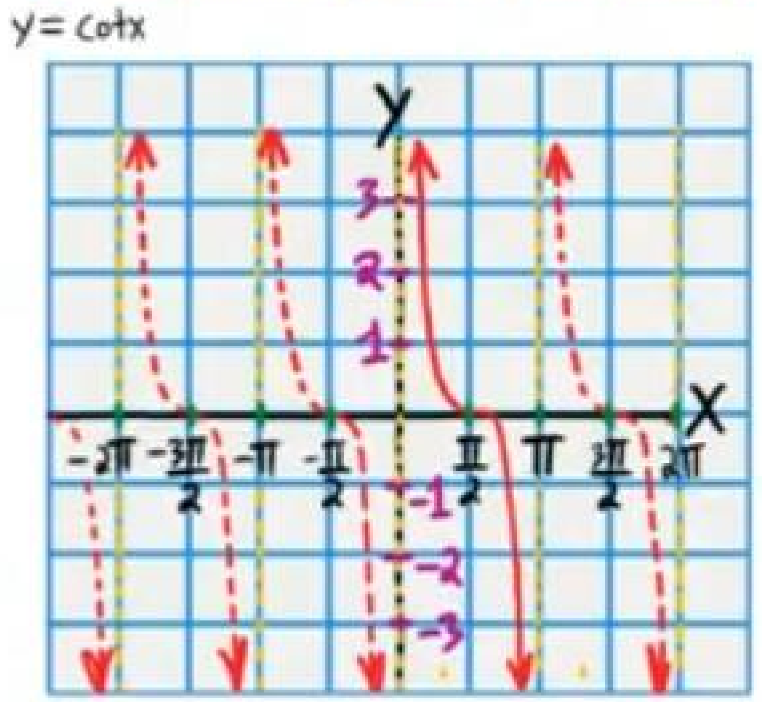
$y = \csc x$



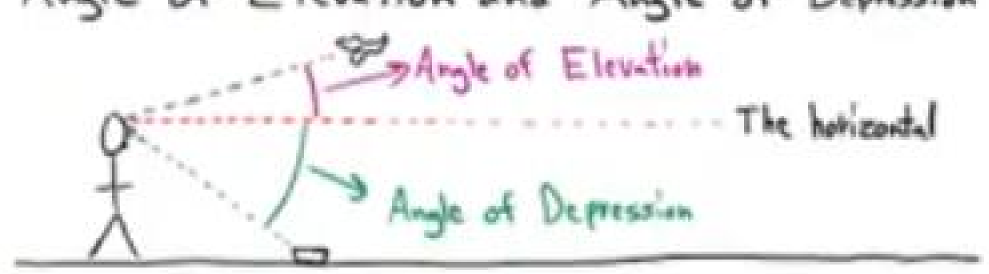
$y = \sec x$



$y = \cot x$



Angle of Elevation and Angle of Depression



Converting from Degrees to Radians

degree measure $\times \frac{\pi \text{ radians}}{180 \text{ degrees}}$ = radian measure

Arc Length

$\boxed{s = \theta r}$ *This formula can be applied with radian units only

The Area of a Sector

$\boxed{A = \frac{\theta}{2}r^2}$ *This formula can be applied with radian units only.

Converting from Radians to Degrees

Radian measure $\times \frac{180 \text{ degrees}}{\pi \text{ radians}}$ = degree measure

General Forms of Trigonometric Functions

Sinusoidal functions:

$y = A\sin(\omega x - \phi) + B$

$y = A\cos(\omega x - \phi) + B$

$y = A\csc(\omega x - \phi) + B$ $\quad y = A\tan(\omega x - \phi) + B$

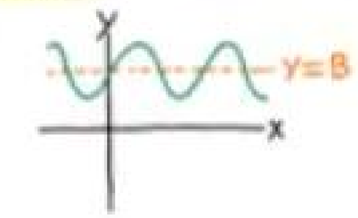
$y = A\sec(\omega x - \phi) + B$ $\quad y = A\cot(\omega x - \phi) + B$

Sinusoidal Function: A function whose graph has the same general form as the sine or cosine functions.

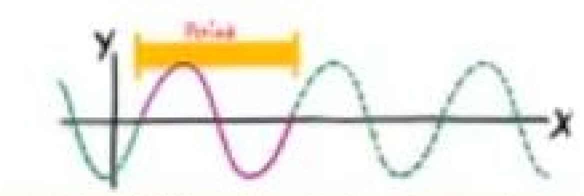
Midline: A horizontal line drawn through the middle of the graph of a trigonometric function to assist when graphing.

For the functions above, the equation of the midline is $y = B$.

Example:



Period: The number of degrees required for a trigonometric function to complete one whole cycle



The period of the sine, cosine, cosecant, and secant functions is given by the following equation.

$P = \frac{2\pi}{\omega}$

The period of the tangent and cotangent functions is given by the following equation.

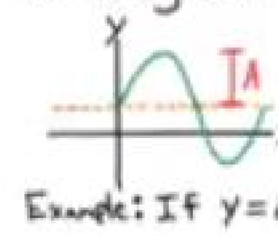
$P = \frac{\pi}{\omega}$

Phase Shift: The horizontal distance that a parent graph is translated (to the right or left).

The phase shift is given by the equation below.

$PS = \frac{\phi}{\omega}$

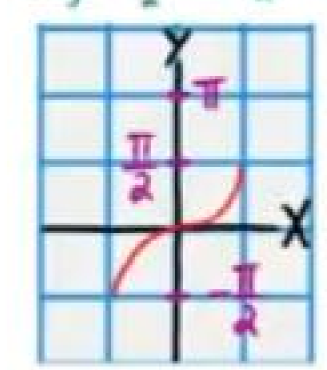
Amplitude: The vertical distance from the midline to the highest or lowest point on a sinusoidal graph



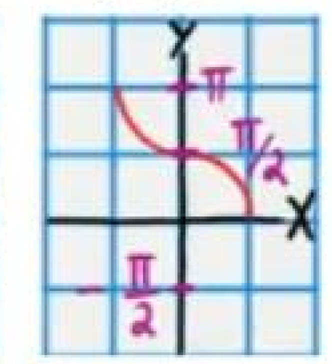
The amplitude is the magnitude (i.e. the absolute value) of the coefficient of the trigonometric function.

Example: If $y = A\sin(\omega x - \phi) + B$, then Amplitude $= |A|$.

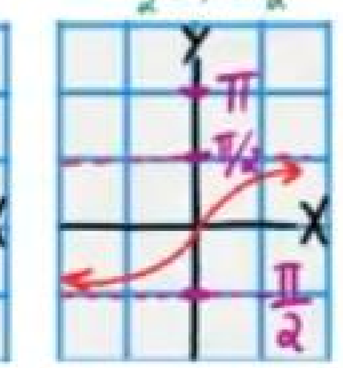
$y = \sin^{-1}x$ $\quad$ Range: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$



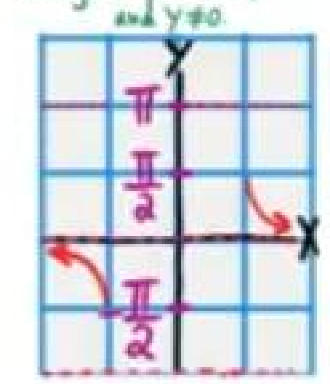
$y = \cos^{-1}x$ $\quad 0 \le y \le \pi$



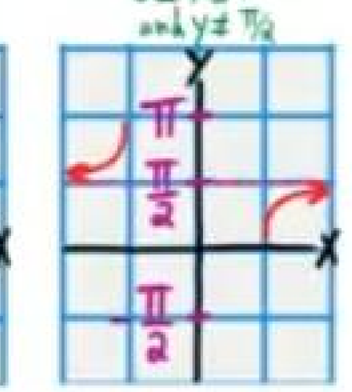
$y = \tan^{-1}x$ $\quad -\frac{\pi}{2} < y < \frac{\pi}{2}$



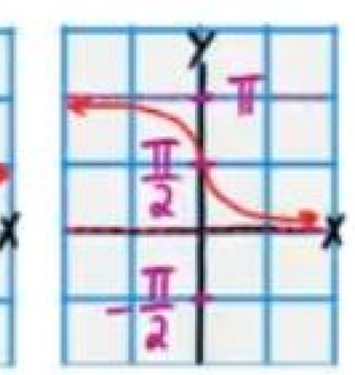
$y = \csc^{-1}x$ $\quad$ Range: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$ and $y \ne 0$



$y = \sec^{-1}x$ $\quad 0 \le y \le \pi$ and $y \ne \pi/2$



$y = \cot^{-1}x$ $\quad 0 < y < \pi$



If an object moves s units on a circular path in t units of time, then V is defined as the linear velocity according to the equation below.

$V = \frac{s}{t}$

If an object on a circle rotates θ angle units in t units of time, then ω is defined as the angular velocity according to the equation below.

$\omega = \frac{\theta}{t}$

$\boxed{V = r\omega}$ *This formula can be applied with radian units only.

Revolution: A complete rotation.

Revolutions per minute = rpm

1 Revolution = 2π radians = 360°

④ $y=\sqrt{1+2x^2}$

$$y=(1+2x^2)^{\frac{1}{2}}$$

$$\frac{dy}{dx}=\frac{1}{2}(1+2x^2)^{\frac{1}{2}-1}\frac{d}{dx}(1+2x^2)$$

$$\frac{dy}{dx}=\frac{1}{2}(1+2x^2)^{-\frac{1}{2}}(4x)$$

$$\frac{dy}{dx}=\frac{1}{2}\cdot\frac{1}{(1+2x^2)^{1/2}}(4x)$$

$$\frac{dy}{dx}=\frac{1}{2}\cdot\frac{1}{\sqrt{1+2x^2}}(4x)$$

$$\frac{dy}{dx}=\frac{4x}{2\sqrt{1+2x^2}}$$

$$\frac{dy}{dx}=\boxed{\frac{2x}{\sqrt{1+2x^2}}}$$

⑤ $y=(5x+1)^5$

$$\frac{dy}{dx}=5(5x+1)^{5-1}\frac{d}{dx}(5x+1)=5(5x+1)^4(5)$$

$$=\boxed{25(5x+1)^4}$$

Find y'.

⑥ $y=\tan(-6x^4)$

$$y'=\sec^2(-6x^4)\frac{d}{dx}(-6x^4)=\left[\sec^2(-6x^4)\right](-24x^3)$$

$$=\boxed{-24x^3\sec^2(-6x^4)}$$

(7) $y = \sqrt[3]{1+\cot x}$

$$y = (1+\cot x)^{\frac{1}{3}}$$

$$y' = \frac{1}{3}(1+\cot x)^{\frac{1}{3}-1}\frac{d}{dx}(1+\cot x)$$

$$= \frac{1}{3}(1+\cot x)^{-\frac{2}{3}}(-\csc^2 x)$$

$$= \frac{1}{3}\cdot\frac{1}{(1+\cot x)^{2/3}}(-\csc^2 x)$$

$$= \frac{1}{3}\cdot\frac{1}{\sqrt[3]{(1+\cot x)^2}}(-\csc^2 x)$$

$$= \boxed{-\frac{\csc^2 x}{3\sqrt[3]{(1+\cot x)^2}}}$$

(8) $y = \cos(\csc x)$

$$y' = -\sin(\csc x)\frac{d}{dx}(\csc x)$$

$$y' = [-\sin(\csc x)](-\csc x\cot x)$$

$$y' = \boxed{\csc x\cot x\sin(\csc x)}$$

(9) $y = (-2\sec x+7)^8$

$$y' = 8(-2\sec x+7)^{8-1}\frac{d}{dx}(-2\sec x+7)$$

$$y' = 8(-2\sec x+7)^7(-2\sec x\tan x)$$

$$y' = \boxed{-16\sec x\tan x(-2\sec x+7)^7}$$

(10) $y = \cot(x^4)$

$y' = -\csc^2(x^4)\,\frac{d}{dx}(x^4) = [-\csc^2(x^4)](4x^3)$

$= \boxed{-4x^3\csc^2(x^4)}$

(11) $y = \cot^4 x$

$y = (\cot x)^4$

$y' = 4(\cot x)^{4-1}\frac{d}{dx}(\cot x)$

$y' = 4(\cot x)^3(-\csc^2 x)$

$y' = \boxed{-4\cot^3 x\csc^2 x}$

Find $\frac{dy}{dx}$.

(12) $y = (-6x^3+4x^2-20x+1)^4$

$\frac{dy}{dx} = 4(-6x^3+4x^2-20x+1)^{4-1}\frac{d}{dx}(-6x^3+4x^2-20x+1)$

$\frac{dy}{dx} = \boxed{4(-6x^3+4x^2-20x+1)^3(-18x^2+8x-20)}$

(13) $y = \tan(x^3)$

$\frac{dy}{dx} = \sec^2(x^3)\,\frac{d}{dx}(x^3) = [\sec^2(x^3)](3x^2) = \boxed{3x^2\sec^2(x^3)}$

(14) $y = \tan^3 x$

$y = (\tan x)^3$

$\frac{dy}{dx} = 3(\tan x)^{3-1}\frac{d}{dx}(\tan x) = 3(\tan x)^2\sec^2 x$

$= \boxed{3\tan^2 x\sec^2 x}$

(15) $y = \sqrt[4]{3x^2-2}$

$$y = (3x^2-2)^{\frac{1}{4}}$$

$$\frac{dy}{dx} = \frac{1}{4}(3x^2-2)^{\frac{1}{4}-1}\frac{d}{dx}(3x^2-2)$$

$$= \frac{1}{4}(3x^2-2)^{-\frac{3}{4}}(6x)$$

$$= \frac{1}{4}\cdot\frac{1}{(3x^2-2)^{3/4}}(6x)$$

$$= \frac{1}{4}\cdot\frac{1}{\sqrt[4]{(3x^2-2)^3}}(6x)$$

$$= \frac{6x}{4\sqrt[4]{(3x^2-2)^3}}$$

$$= \boxed{\frac{3x}{2\sqrt[4]{(3x^2-2)^3}}}$$

(16) $y = (6-7x^2)^5$

$$\frac{dy}{dx} = 5(6-7x^2)^{5-1}\frac{d}{dx}(6-7x^2)$$

$$\frac{dy}{dx} = 5(6-7x^2)^4(-14x)$$

$$\frac{dy}{dx} = \boxed{-70x(6-7x^2)^4}$$

Find y'.

(17) $y = \sin(6x)$

$$y' = [\cos(6x)]\frac{d}{dx}(6x)$$

$$y' = [\cos(6x)](6)$$

$$y' = \boxed{6\cos(6x)}$$

(18) $y = \sqrt[5]{\cos x + 8x}$

$y = (\cos x + 8x)^{\frac{1}{5}}$

$y' = \frac{1}{5}(\cos x + 8x)^{\frac{1}{5}-1} \frac{d}{dx}(\cos x + 8x)$

$y' = \frac{1}{5}(\cos x + 8x)^{-\frac{4}{5}}(-\sin x + 8)$

$y' = \frac{1}{5} \cdot \frac{1}{(\cos x + 8x)^{4/5}} (-\sin x + 8)$

$y' = \frac{1}{5} \cdot \frac{1}{\sqrt[5]{(\cos x + 8x)^4}} (-\sin x + 8)$

$y' = \boxed{\frac{-\sin x + 8}{5\sqrt[5]{(\cos x + 8x)^4}}}$

(19) $y = \cot(\sin x)$

$y' = -\csc^2(\sin x) \frac{d}{dx}(\sin x)$

$y' = [-\csc^2(\sin x)](\cos x)$

$y' = \boxed{-\cos x \csc^2(\sin x)}$

(20) $y = (2\pi - \csc x)^3$

$y' = 3(2\pi - \csc x)^{3-1} \frac{d}{dx}(2\pi - \csc x)$

$y' = 3(2\pi - \csc x)^2 [-(-\csc x \cot x)]$

$y' = 3(2\pi - \csc x)^2 \csc x \cot x$

$y' = \boxed{3\csc x \cot x (2\pi - \csc x)^2}$

(21) $y = \cos(x^5)$

$$y' = -\sin(x^5)\,\frac{d}{dx}(x^5) = \left[-\sin(x^5)\right](5x^4)$$

$$= \boxed{-5x^4\sin(x^5)}$$

(22) $y = \cos^5 x$

$$y = (\cos x)^5$$

$$y' = 5(\cos x)^{5-1}\frac{d}{dx}(\cos x) = 5(\cos x)^4(-\sin x)$$

$$= \boxed{-5\cos^4 x\sin x}$$

Find $\frac{dy}{dx}$.

(23) $y = \sec(-x^4+2x-1)$

$$\frac{dy}{dx} = \sec(-x^4+2x-1)\tan(-x^4+2x-1)\,\frac{d}{dx}(-x^4+2x-1)$$

$$= \left[\sec(-x^4+2x-1)\tan(-x^4+2x-1)\right](-4x^3+2)$$

$$= \boxed{(-4x^3+2)\sec(-x^4+2x-1)\tan(-x^4+2x-1)}$$

(24) $y = \csc(\sin x)$

$$\frac{dy}{dx} = -\csc(\sin x)\cot(\sin x)\,\frac{d}{dx}(\sin x)$$

$$= \left[-\csc(\sin x)\cot(\sin x)\right](\cos x)$$

$$= \boxed{-\cos x\csc(\sin x)\cot(\sin x)}$$

(25) $y = \sec(\cos x)$

$$\frac{dy}{dx} = \sec(\cos x)\tan(\cos x)\,\frac{d}{dx}(\cos x)$$

$$\frac{dy}{dx} = \left[\sec(\cos x)\tan(\cos x)\right](-\sin x)$$

$$\frac{dy}{dx} = \boxed{-\sin x\sec(\cos x)\tan(\cos x)}$$

(26) $y = \csc(3x^2+1)$

$$\frac{dy}{dx} = \left[-\csc(3x^2+1)\cot(3x^2+1)\right]\frac{d}{dx}(3x^2+1)$$

$$= \left[-\csc(3x^2+1)\cot(3x^2+1)\right](6x)$$

$$= \boxed{-6x\csc(3x^2+1)\cot(3x^2+1)}$$

Find y'.

(27) $y = -2\cos^3 x$

$y = -2(\cos x)^3$

$y' = -2 \cdot 3 (\cos x)^{3-1} \frac{d}{dx}(\cos x) = -6(\cos x)^2(-\sin x)$

$= \boxed{6 \sin x \cos^2 x}$

(28) $y = 4(2x-1)^4$

$y' = 4 \cdot 4 (2x-1)^{4-1} \frac{d}{dx}(2x-1)$

$y' = 16(2x-1)^3 (2)$

$y' = \boxed{32(2x-1)^3}$

Using the Chain Rule Twice

(29) $y = (x^3-10x-1)^3 + \tan^2 x$

$y = (x^3-10x-1)^3 + (\tan x)^2$

$y' = 3(x^3-10x-1)^2 \frac{d}{dx}(x^3-10x-1) + 2(\tan x)^1 \frac{d}{dx}(\tan x)$

$y' = \boxed{3(x^3-10x-1)^2(3x^2-10) + 2\tan x \sec^2 x}$

(30) $y = \sin(3x) - (2x^2+x-8)^5$

$y' = \cos(3x) \frac{d}{dx}(3x) - 5(2x^2+x-8)^4 \frac{d}{dx}(2x^2+x-8)$

$y' = [\cos(3x)](3) - 5(2x^2+x-8)^4(4x+1)$

$y' = \boxed{3\cos(3x) - (20x+5)(2x^2+x-8)^4}$

A Function within a Function within a Function

If $r(x) = f(g(h(x)))$,

then $r'(x) = f'(g(h(x)))\, g'(h(x))\, h'(x)$

(31) $y = \cot(\sin(5x))$

$$y' = -\csc^2(\sin(5x)) \frac{d}{dx}(\sin(5x))$$

$$y' = [-\csc^2(\sin(5x))] \cos(5x) \frac{d}{dx}(5x)$$

$$y' = [-\csc^2(\sin(5x))][\cos(5x)](5)$$

$$y' = \boxed{-5\cos(5x)\csc^2(\sin(5x))}$$

(32) $y = (4 - 3\cos^2 x)^3$

$$y = (4 - 3(\cos x)^2)^3$$

$$y' = 3(4 - 3(\cos x)^2)^{3-1} \frac{d}{dx}(4 - 3(\cos x)^2)$$

$$y' = 3(4 - 3\cos^2 x)^2 \left(-3 \cdot 2(\cos x)^{2-1} \frac{d}{dx}(\cos x)\right)$$

$$y' = 3(4 - 3\cos^2 x)^2 (-6(\cos x)^1(-\sin x))$$

$$y' = 3(4 - 3\cos^2 x)^2 (6\sin x \cos x)$$

$$y' = \boxed{18\sin x \cos x (4 - 3\cos^2 x)^2}$$

Find $\frac{dy}{dx}$.

(33) $y = \sqrt[7]{2(4x-1)^5+10}$

$$y = \left(2(4x-1)^5+10\right)^{\frac{1}{7}}$$

$$\frac{dy}{dx} = \frac{1}{7}\left(2(4x-1)^5+10\right)^{\frac{1}{7}-1}\frac{d}{dx}\left(2(4x-1)^5+10\right)$$

$$\frac{dy}{dx} = \frac{1}{7}\left(2(4x-1)^5+10\right)^{-\frac{6}{7}}\left(2\cdot 5(4x-1)^4\frac{d}{dx}(4x-1)\right)$$

$$= \frac{1}{7}\left(2(4x-1)^5+10\right)^{-\frac{6}{7}}\left(10(4x-1)^4(4)\right)$$

$$= \boxed{\frac{40}{7}\frac{(4x-1)^4}{\sqrt[7]{\left(2(4x-1)^5+10\right)^6}}}$$

(34) $y = \left(7+\sqrt{-x+4}\right)^3$

$$y = \left(7+(-x+4)^{\frac{1}{2}}\right)^3$$

$$\frac{dy}{dx} = 3\left(7+(-x+4)^{\frac{1}{2}}\right)^2\frac{d}{dx}\left(7+(-x+4)^{1/2}\right)$$

$$\frac{dy}{dx} = 3\left(7+(-x+4)^{1/2}\right)^2\left(\frac{1}{2}(-x+4)^{\frac{1}{2}-1}\frac{d}{dx}(-x+4)\right)$$

$$\frac{dy}{dx} = 3\left(7+(-x+4)^{1/2}\right)^2\left(\frac{1}{2}(-x+4)^{\frac{1}{2}-1}(-1)\right)$$

$$\frac{dy}{dx} = -\frac{3}{2}\left(7+\sqrt{-x+4}\right)^2(-x+4)^{-\frac{1}{2}}$$

$$\frac{dy}{dx} = -\frac{3}{2}\left(7+\sqrt{-x+4}\right)^2\frac{1}{\sqrt{-x+4}}$$

$$\frac{dy}{dx} = \boxed{-\frac{3\left(7+\sqrt{-x+4}\right)^2}{2\sqrt{-x+4}}}$$

(35) $y = \sec^2\left(\frac{x}{2}\right)$

$$y = \left(\sec\left(\frac{x}{2}\right)\right)^2$$

$$\frac{dy}{dx} = 2\left(\sec\left(\frac{x}{2}\right)\right)^1 \frac{d}{dx}\sec\left(\frac{x}{2}\right)$$

$$\frac{dy}{dx} = 2\sec\left(\frac{x}{2}\right) \sec\left(\frac{x}{2}\right)\tan\left(\frac{x}{2}\right) \left(\frac{1}{2}\right)$$

$$\boxed{\frac{dy}{dx} = \sec^2\left(\frac{x}{2}\right)\tan\left(\frac{x}{2}\right)}$$

(36) $y = \sin^3(x^2)$

$$y = \left(\sin(x^2)\right)^3$$

$$\frac{dy}{dx} = 3\left(\sin(x^2)\right)^2 \frac{d}{dx}\left(\sin(x^2)\right)$$

$$\frac{dy}{dx} = \left[3\sin^2(x^2)\right] \cos(x^2) \frac{d}{dx}(x^2)$$

$$\frac{dy}{dx} = \left[3\sin^2(x^2)\cos(x^2)\right](2x)$$

$$\frac{dy}{dx} = \boxed{6x\cos(x^2)\sin^2(x^2)}$$

(37) $y = \cot(\cot(x^3))$

$$\frac{dy}{dx} = -\csc^2(\cot(x^3)) \frac{d}{dx}(\cot(x^3))$$

$$\frac{dy}{dx} = -\csc^2(\cot(x^3))\left[-\csc^2(x^3)\right]\frac{d}{dx}(x^3)$$

$$\frac{dy}{dx} = -\csc^2(\cot(x^3))\left[-\csc^2(x^3)\right](3x^2)$$

$$\frac{dy}{dx} = \boxed{3x^2\csc^2(x^3)\csc^2(\cot(x^3))}$$

(38) $y = \sin(\sin(\sin x))$

$$\frac{dy}{dx} = \cos(\sin(\sin x)) \frac{d}{dx}(\sin(\sin x))$$

$$\frac{dy}{dx} = \left[\cos(\sin(\sin x))\right]\cos(\sin x)\frac{d}{dx}(\sin x)$$

$$\frac{dy}{dx} = \left[\cos(\sin(\sin x))\right]\left[\cos(\sin x)\right](\cos x)$$

$$\frac{dy}{dx} = \boxed{\cos x\cos(\sin x)\cos(\sin(\sin x))}$$

Using the Chain Rule Three Times

A Function within a Function within a Function within a Function

Let $z(x) = f(g(h(v(x))))$

Then $z'(x) = f'(g(h(v(x))))g'(h(v(x)))h'(v(x))v'(x)$

Find y'

(39) $y = \cos(\tan^2(-\frac{x^4}{2}))$

$$y = \cos\left[\left(\tan\left(-\frac{x^4}{2}\right)\right)^2\right]$$

$$y' = -\sin\left[\left(\tan\left(-\frac{x^4}{2}\right)\right)^2\right]\frac{d}{dx}\left[\left(\tan\left(-\frac{x^4}{2}\right)\right)^2\right]$$

$$y' = -\sin\left[\left(\tan\left(-\frac{x^4}{2}\right)\right)^2\right] 2\left[\tan\left(-\frac{x^4}{2}\right)\right]^{2-1}\frac{d}{dx}\tan\left(-\frac{x^4}{2}\right)$$

$$y' = -\sin\left[\left(\tan\left(-\frac{x^4}{2}\right)\right)^2\right] 2\left[\tan\left(-\frac{x^4}{2}\right)\right]^1 \sec^2\left(-\frac{x^4}{2}\right)\frac{d}{dx}\left(-\frac{x^4}{2}\right)$$

$$y' = -\sin\left[\left(\tan\left(-\frac{x^4}{2}\right)\right)^2\right] 2\left[\tan\left(-\frac{x^4}{2}\right)\right]\left[\sec^2\left(-\frac{1}{2}x^4\right)\right](-2x^3)$$

$$y' = \boxed{4x^3\tan\left(-\frac{x^4}{2}\right)\sec^2\left(-\frac{x^4}{2}\right)\sin\left[\tan^2\left(-\frac{x^4}{2}\right)\right]}$$

(40) $y = \cot(\cos(\cos(x^2)))$

$$y' = -\csc^2(\cos(\cos(x^2)))\frac{d}{dx}(\cos(\cos(x^2)))$$

$$y' = \left[-\csc^2(\cos(\cos(x^2)))\right](-\sin(\cos(x^2)))\frac{d}{dx}(\cos(x^2))$$

$$y' = \left[-\csc^2(\cos(\cos(x^2)))\right]\left[-\sin(\cos(x^2))\right](-\sin(x^2))\frac{d}{dx}(x^2)$$

$$y' = \left[-\csc^2(\cos(\cos(x^2)))\right]\left[-\sin(\cos(x^2))\right]\left[-\sin(x^2)\right](2x)$$

$$y' = \boxed{-2x\sin(x^2)\sin(\cos(x^2))\csc^2(\cos(\cos(x^2)))}$$

A Proof of the Chain Rule

Let $r(x) = f(g(x))$

Then $r'(x) = \lim_{h \to 0} \frac{r(x+h) - r(x)}{h}$

$$= \lim_{h \to 0} \frac{f(g(x+h)) - f(g(x))}{h}$$

$$= \lim_{h \to 0} \left[\frac{f(g(x+h)) - f(g(x))}{h} \cdot \frac{g(x+h) - g(x)}{g(x+h) - g(x)} \right]$$

$$= \lim_{h \to 0} \left[\frac{f(g(x+h)) - f(g(x))}{g(x+h) - g(x)} \cdot \frac{g(x+h) - g(x)}{h} \right]$$

$$\stackrel{3}{=} \lim_{h \to 0} \frac{f(g(x+h)) - f(g(x))}{g(x+h) - g(x)} \lim_{h \to 0} \frac{g(x+h) - g(x)}{h}$$

$$= f'(g(x))\, g'(x)$$